

Comprehensive STLC Study Guide with Diagrams, Examples & Interview Prep

✅ 1. STLC Overview

Summary from Transcript:

STLC (Software Testing Life Cycle) is a sequence of activities performed during testing to ensure software quality.

Additional Details:

Phases:

- Requirement Analysis
- Test Planning
- Test Case Development
- Test Environment Setup
- Test Execution
- Test Closure

Diagram:

[Requirement Analysis] → [Test Planning] → [Test Design] → [Test Execution] → [Defect Reporting] → [Test Closure]

Real-world Example:

A banking app undergoes STLC where testers analyze requirements, plan tests, write cases for login, execute them, report bugs, and close testing after fixes.

Interview Importance:

Very common question for QA roles.

Possible Interview Questions:

- What is STLC?
 - Explain STLC phases with examples.
 - Draw STLC workflow.
 - Why is STLC important?
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✓ 2. SDLC vs STLC

Summary from Transcript:

SDLC = Development process; STLC = Testing process.

Diagram:

SDLC: Requirements → Design → Development → Testing → Deployment → Maintenance

STLC: Requirements → Test Planning → Test Design → Test Execution → Closure

Example:

In SDLC, developers code features; in STLC, testers validate those features.

Interview Questions:

- Differentiate SDLC and STLC.
 - How do STLC phases map to SDLC?
 - Why do we need STLC?
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✓ 3. Test Planning

Summary:

Defines scope, approach, resources, and schedule for testing.

Best Practices:

- Define scope clearly
- Estimate time and cost accurately
- Review plan with stakeholders

Diagram:

Inputs: Requirements → Activities: Scope, Resources, Schedule → Output: Test Plan Document

Example:

For an e-commerce app, test planning includes deciding to test payment gateway, assigning 2 testers, and scheduling 2 weeks.

Interview Questions:

- What is test planning?
- What does a test plan contain?
- How do you estimate effort?

✓ 4. Test Design / Test Case Development

Summary:

Writing test cases and preparing test data.

Best Practices:

- Cover positive & negative scenarios
- Maintain traceability matrix

Example Test Case (Login):

Test Case ID	Steps	Expected Result
TC001	Enter valid username & password	Login successful

Diagram:

Requirements → Test Scenarios → Test Cases → Test Data

Interview Questions:

- How do you write a test case?
- Difference between test case & scenario?
- Write a test case for login.

✓ 5. Test Execution

Summary:

Executing test cases and logging results.

Example:

Run TC001 for login → Pass → Update status in test log.

Diagram:

Test Cases → Execution → Status (Pass/Fail) → Defect Logging

Interview Questions:

- What is test execution?
- How do you track progress?
- Explain regression testing.

✅ 6. Defect Reporting & Tracking

Summary:

Documenting and tracking bugs until closure.

Defect Lifecycle Diagram:

New → Assigned → Fixed → Retested → Closed

Example Defect Report:

- **ID:** BUG123
- **Summary:** Login fails with valid credentials
- **Steps:** Enter valid username/password → Click Login
- **Expected:** Login successful
- **Actual:** Error message displayed

Interview Questions:

- Explain defect lifecycle.
- What details should a defect report contain?
- Which tools have you used?

✅ 7. Test Closure

Summary:

Final phase with reports and lessons learned.

Activities:

- Test summary report
- Metrics analysis
- Archive artifacts

Diagram:

Execution Completed → Reports → Sign-off

Interview Questions:

- What is test closure?

- Why is it important?
 - What does a closure report include?
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✅ 8. Exit Criteria

Summary:

Conditions to stop testing.

Examples:

- All test cases executed
- No critical defects remain
- Coverage $\geq 95\%$

Diagram:

Check Coverage → Check Defects → Approve Closure

Interview Questions:

- What are exit criteria?
 - Give an example for a web app.
 - Why are they important?
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✅ Quick Revision Table for Interview Prep

Topic	Key Points	Common Questions
STLC Overview	Phases of testing	What is STLC? Explain phases.
SDLC vs STLC	Development vs Testing lifecycle	Differentiate SDLC & STLC.
Test Planning	Scope, resources, schedule	What does a test plan contain?
Test Design	Writing test cases	Write a test case for login.
Test Execution	Running tests, logging results	How do you track execution?
Defect Reporting	Bug lifecycle, tools	Explain defect lifecycle.
Test Closure	Reports, metrics, sign-off	Why is closure important?

Topic**Key Points****Common Questions**

Exit Criteria

Conditions to stop testing

What are exit criteria?